MCA/BCA STUDENT DISSERTATION / CAPSTONE ETHICAL CLEARANCE FORM

Department: PG Dept. of Computer Applications | Academic Year: 2025-2026

A. STUDENT & PROJECT DETAILS

1. Student Name	**Athulya** *(or enter your full registered name)*
2. Admission Number	**25PMC_____** *(fill in your registered number)*
3. Programme of Study	☑ **MCA** ☐ BCA
4. Institution Email	athulya3404@gmail.com *(or student institutional email)*
5. Contact Phone	[Student Contact Phone Number]
6. Project Title	**EasyTech – Full-Stack Electronic Gadgets E-Commerce System with AI-Powered Assistance & Secure Transaction Lifecycle**
7. Project Modality	☐ Dissertation ☐ Capstone ☐ Internship Project ☑ **Other: Mini Project**
8. Supervisor (Faculty Guide)	[Faculty Guide Name]
9. DRAC / Dept. Advisory Member	[Member Name, if applicable]
10. Industry / External Mentor	N/A
11. Estimated Start Date	30 July 2025
12. Estimated Completion Date	30 October 2025

B. PROJECT OVERVIEW

13. Purpose & Rationale	The aim of this project, **EasyTech (EasyTechStore)**, is to develop an enterprise-grade, full-stack e-commerce web platform engineered for purchasing electronic gadgets (smartphones, laptops, audio equipment, computer accessories) with intelligent AI assistance and robust transaction integrity. It addresses the real-world problems of high-concurrency race conditions during flash sales that lead to inventory overselling, security vulnerabilities in client-side payment processing, the difficulty customers face when comparing complex multi-dimensional hardware specifications (CPU, RAM, display, battery), and the lack of personalized, context-aware technical product advice. The proposed system provides a centralized platform where customers can browse catalog items with faceted filtering, perform side-by-side technical gadget comparisons, reserve stock with atomic database-level locks, execute secure simulated payments with cryptographic HMAC-SHA256 signature verification, track order lifecycles (Pending, Paid, Shipped, Delivered), and download GST-compliant PDF invoices. Store managers can oversee catalog CRUD operations, perform batch Excel/CSV imports, manage order state transitions, and monitor sales analytics. Additionally, the system incorporates a Retrieval-

	Augmented Generation (RAG) AI Chatbot powered by ChromaDB vector embeddings and Large Language Models (LLM) to assist shoppers with natural language gadget inquiries, budget filtering, and technical compatibility advice. The expected contribution of this project is to eliminate inventory leakage under concurrent traffic, ensure tamper-proof payment verification, streamline complex hardware evaluation for buyers, and demonstrate a secure, production-ready e-commerce architecture following clean software engineering standards.
14. Methodology / Technical Approach	☑ **Software/System Development Model** ☐ Data Analytics/Mining ☑ **AI/ML/DL Development** ☐ HCI/Usability Testing ☐ Survey/Interviews ☐ Secondary Data Analysis ☐ Network Measurement ☐ IoT/Robotics ☐ Web Scraping/Crawling ☐ Security/Penetration Testing (with written authorization only) **Brief description (design, datasets, tools, procedures, evaluation):** EasyTech (EasyTechStore) is a web-based e-commerce platform developed using Python (Django Framework), Django REST Framework, HTML5, CSS3, Bootstrap 5, JavaScript, Redis, and PostgreSQL for database management. The system automates electronic gadget discovery, inventory reservation, cryptographic payment validation, order fulfillment, and AI-driven customer support through a unified portal for customers and store managers. Customers can explore products with faceted filters (category, brand, price range), compare gadget specifications side by side, manage shopping carts with real-time stock verification, and track order lifecycles with automated ReportLab PDF invoice generation. Store managers can manage inventory, process batch catalog updates via Excel/CSV (OpenPyXL), and fulfill orders through a finite state machine. The system incorporates Artificial Intelligence and Retrieval-Augmented Generation (RAG) using ChromaDB vector database and LLMs (Groq / Ollama Llama 3.2 / Qwen) to deliver natural language product recommendations and technical assistance grounded on catalog data. It follows a modular Model–View–Template (MVT) architecture with decoupled domain applications (users, products, category, cart, orders, chatbot) to ensure high maintainability, testability, and scalability. Concurrency controls employ pessimistic row-level locking (`select_for_update`) and atomic SQL `F()` expressions to prevent race conditions during checkout, coupled with a 15-minute order reservation expiration worker (`cleanup_expired_orders`). Two-stage payment processing uses the Strategy/Adapter pattern with HMAC-SHA256 signature verification (`hmac.compare_digest`) for payment integrity. Evaluation focuses on automated unit and integration tests using Pytest, concurrency load testing verifying zero stock overselling, cryptographic tamper testing, and vector retrieval accuracy across a 1,000-question benchmark dataset.
15. Software / Hardware & Infrastructure	**Primary languages/frameworks:** Python, Django Framework, Cookiecutter Django (project scaffolding), Django REST Framework, HTML5, CSS3, Bootstrap 5, JavaScript, PostgreSQL, Redis **External libraries/models (names & licenses):** Django (BSD License), Bootstrap (MIT License), Django REST Framework (BSD License), ChromaDB (Apache-2.0 License), OpenAI Python SDK (Apache-2.0 License), ReportLab (BSD License), OpenPyXL (MIT License), Argon2-cffi (MIT License), Django-Allauth (MIT License), Psycopg (PostgreSQL/LGPL License), Redis-py (MIT License), and Gunicorn (MIT License). LLM Models: Llama 3.2 (Meta Community License) via Ollama, and Groq Cloud API (Commercial API terms for development). **Compute/Cloud services (provider, region, data location):** Local development environment using Docker and Docker Compose for PostgreSQL, Redis, and Traefik deployment. Amazon Web Services (AWS) – EC2 for application deployment with

	Traefik reverse proxy and Let's Encrypt SSL (https://easystore1.me). Amazon Web Services (AWS) – EC2, EU North (Stockholm) / US East, Data hosted in AWS Cloud. **On-prem resources/labs:** Personal computer with Python, Django, PostgreSQL, Docker Desktop, Visual Studio Code, Git, GitHub, and internet connectivity.

C. DATA, PARTICIPANTS, AND ETHICS

16. Human Participants Involvement	☐ Yes ☑ **No** *(If Yes, answer Q17–Q24. If No, skip to Q25.)*
25. Datasets & Digital Sources	**Primary datasets/sources (name, URL/DOI, owner):** Custom sample data (synthetic electronic gadgets catalog comprising smartphones, laptops, audio equipment, dummy prices, SKUs, and specifications), and a curated benchmark suite of 1,000 domain test queries for AI chatbot evaluation. **Licenses/Terms of Use (attach evidence of permissible use):** N/A **Provenance & Documentation (dataset cards/datasheets, model cards):** N/A
26. Web Scraping / Crawling	Target domains/URLs: N/A Respect for robots.txt and rate-limits: ☐ Yes ☑ **N/A** Terms of Service compliance & permission letters (attach): ☐ Attached ☑ **N/A** Data minimization & exclusion of personal data: ☑ **Ensured** Anti-disruption safeguards (throttling, off-peak scheduling): ☑ **Implemented / N/A**
27. Security / Penetration Testing	Scope and assets under explicit written authorization (attach proof): ☐ Attached ☑ **N/A** Non-interference & logging plan; rollback procedures: Testing is restricted exclusively to local mock sandbox environments and Docker loopback interfaces; no external networks, payment gateways, or third-party production systems are targeted. Legal and institutional policy compliance statement: All security checks (CSRF verification, SQL injection protection via Django ORM, and HMAC signature tamper tests) strictly comply with institutional IT usage policies and academic ethics.

D. DATA PROTECTION & MANAGEMENT (ALL PROJECTS)

28. Personal Data Handling	☑ **No personal data processed** ☐ Pseudonymized data only ☐ Personal data (contact info, identifiers) ☐ Special categories/sensitive data *(All customer profiles, shipping addresses, phone numbers, and payment tokens in the system are synthetic dummy records generated purely for academic testing.)*
29. Data Protection Compliance	Applicable laws/policies (institutional policy, national data protection law, platform TOS): Institutional policy, Information Technology (IT) Act (India), OWASP Top 10 Web Application Security guidelines Legal basis/permission for processing (consent, license, contractual permission, legitimate use): Legitimate educational use and academic software development
30. Data Lifecycle Plan	Storage location (drive/server/cloud & region): Local File System, local PostgreSQL container, and AWS EC2 cloud instance (EU North / Stockholm). Security (encryption, access control, keys/passwords): Argon2 password hashing, role-based access control (RBAC), Django CSRF tokens, environment variable isolation (`.env` files) for `SECRET_KEY` and API keys, and constant-time HMAC-SHA256 signature verification (`hmac.compare_digest`). Retention period: 4 months | Destruction method & date: Secure deletion after academic completion and degree evaluation, 15/11/2026

	Sharing (who will access; will de-identified data/code be released?): ☑ **Yes** ☐ No If Yes, repository/license and de-identification approach: https://github.com/athulya3404/EasyTechStore (Released under MIT License; all sensitive credentials, secret keys, and private `.env` configurations are excluded from the repository via `.gitignore`).
31. Bias, Fairness & Safety (for AI/ML/NLP/CV)	**Expected risks (bias, representational harms, unsafe outputs):** The RAG-based AI chatbot may produce hallucinations, out-of-catalog product recommendations, or inaccurate hardware compatibility advice when responding to complex user queries. **Mitigations (sampling, audits, human oversight, content filters, red-teaming):** Use Retrieval-Augmented Generation (RAG) strictly grounded on verified store inventory and technical spec sheets stored in the ChromaDB vector database. System prompts enforce explicit guardrails requiring the assistant to decline non-electronics queries and acknowledge when a product or feature is not in the store catalog. Evaluated across 1,000 diverse test questions to verify intent extraction and price-filtering accuracy. **Model reporting (model card/limitations/appropriate use):** Document the AI/RAG architecture, vector retrieval parameters, and evaluation benchmark in the project report. The chatbot is designed as an advisory shopping aid and customer support assistant, not an autonomous transactional decision-maker.
32. Intellectual Property & Licensing	Ownership of code/data/models: Own Open-source release plan and license (e.g., MIT, Apache-2.0, CC-BY): MIT License Third-party license compliance confirmed: ☑ **Yes**
33. Environmental & Cost Considerations	Estimated compute (GPU hours/energy) and minimization steps: N/A (Standard CPU workstation and lightweight cloud instance; local ChromaDB embeddings and cached queries minimize API calls and compute overhead). Budget/credits and cost controls: Free-tier API keys, open-source software stack, and low-cost/free-tier AWS EC2 cloud instance; minimal financial expenditure.

E. TRAINING, CONFLICTS, AND SAFETY

34. Research Ethics / Compliance Training	☑ **Yes** (course/date): 01/10/2026 ☐ No (commitment date: _____)
35. Conflicts of Interest	☑ **None** ☐ Disclosed below: _____
36. Health & Safety	☐ Yes (attach) ☑ **Not applicable**

F. DECLARATIONS

Student Declaration

I, **Athulya**, affirm that the information provided is accurate and complete. I understand that approval is required before commencing any data collection, scraping, security testing, or human-participant activities; that unauthorized testing or data acquisition is prohibited; and that any substantive change in design, methods, data sources, or risk profile requires re-submission for approval.

Supervisor (Faculty Guide) Approval

I, _____, have reviewed the proposal and ethics safeguards with the student. I verify that the project's scope, data handling, and risk mitigations are appropriate for MCA standards and institutional policies.

Signature (Supervisor): _____

Date: 31 OCTOBER 2025

Signature (Student): _____ **Date: 31 OCTOBER 2025**	
DRAC / Department Member Review Name: _____ **Signature:** _____ **Date: 31 OCTOBER 2025**	**Industry / External Mentor Acknowledgement** Name & Org: N/A **Signature:** _____ **Date: _____**

G. SUBMISSION & PROCESSING

Please submit the signed form with all attachments (study instruments, consent forms, permissions, dataset licenses/TOS evidence, risk assessments, authorization letters for testing/scraping) to the Department Ethics Committee—Computer Applications (MCA/BCA).
Processing time: Please allow at least four weeks for review.

H. OFFICE USE ONLY — DEPARTMENT ETHICS COMMITTEE (MCA/BCA)

Chair/Reviewer: _____	**Date Received: _____** **Review Date: _____**
Decision: ☐ Approved ☐ Minor Revisions ☐ Major Revisions ☐ Not Approved	
Conditions / Comments: _____ _____	
Follow-Up Required By (date): _____	**Final Sign-off (Chair):** _____ **Date: _____**

Attachments Checklist (tick if attached):

☑ **Project synopsis/abstract**	☐ Recruitment messages (if any)
☑ **Detailed methodology / system design**	☑ **Data management & retention plan**
☐ Consent forms & participant materials (if any)	☐ Risk assessment (lab/IoT/robotics/network)
☑ **Dataset documentation & licenses / TOS**	☑ **Ethics/compliance training proof**
☐ Web-scraping plan & permissions (if applicable)	☐ Security testing authorization (if applicable)